LLM Response Collection Method for the StackEval Benchmark

Supporting research note for Decision Log 3

This document summarizes the reasoning behind the choice between manual prompting and API-based querying for collecting LLM responses in the recreated StackEval benchmark. It consolidates the main comparison points, practical implications, and evidence used during the research process.

1. Project context

Project: Recreating StackEval – Real-World LLM Benchmark

Goal: Collect real programming questions, obtain responses from selected LLMs, and later let developers score those responses on a 0–3 scale.

Selected models: GPT-4o, Claude 3.5 Sonnet, Llama 3.1, and GPT-4o mini.

2. Research question

Should the StackEval benchmark use API-based querying or manual prompting to collect LLM responses for real programming questions?

3. Why this decision matters

- Every model should receive the same question under the same conditions.
- The benchmark should remain manageable when multiple questions and multiple models are involved.
- The response collection process should be documented clearly enough to be repeated later.

4. Comparison of the two approaches

Comparison point	Manual prompting	API-based querying
Ease of starting	Very easy to start; no coding is required.	Requires setup such as API keys, scripts, and request handling.
Consistency	Higher risk of prompt variation, copy-paste mistakes, and accidental differences between models.	Supports identical prompt structure and controlled querying conditions across all models.
Automation	Each question and answer cycle must be handled manually.	Questions can be sent automatically and answers can be saved directly.

Comparison point	Manual prompting	API-based querying
Scalability	Becomes repetitive and inefficient as the number of questions grows.	More suitable for larger batches of questions and repeated runs.
Reproducibility	Difficult to guarantee the exact same workflow later.	The same script and dataset structure can be reused for repeatable experiments.
Professional relevance	Useful for quick exploration only.	More aligned with benchmark methodology and future GenAI engineering practice.

5. Main pros and cons

Manual prompting	API-based querying
• Pros: simple, immediate, no setup required. • Pros: useful for very small exploratory tests. • Cons: introduces more human error and inconsistency. • Cons: repetitive and less scalable. • Cons: weaker fit for a structured benchmark.	• Pros: consistent input conditions across models. • Pros: supports automation and systematic data collection. • Pros: easier to rerun in a reproducible way. • Pros: more aligned with professional benchmark workflows. • Cons: requires technical setup and API access.

6. Reasoned conclusion

Manual prompting is workable for a very small informal comparison, but it becomes weaker when the goal is to run a benchmark that should be fair, structured, and repeatable.

API-based querying is more suitable because it supports three core requirements at the same time: consistency, automation, and reproducibility.

For this project, the final decision was to use API-based querying instead of manual prompting to collect LLM responses.

7. Evidence used

- StackEval GitHub repository: https://github.com/ProsusAI/stack-eval
- Socratic dialogue used to refine the reasoning: ChatGPT shared conversation
- Supporting project documents reviewed during the analysis included the project understanding document, the model selection decision log, and the API-vs-manual decision log.